

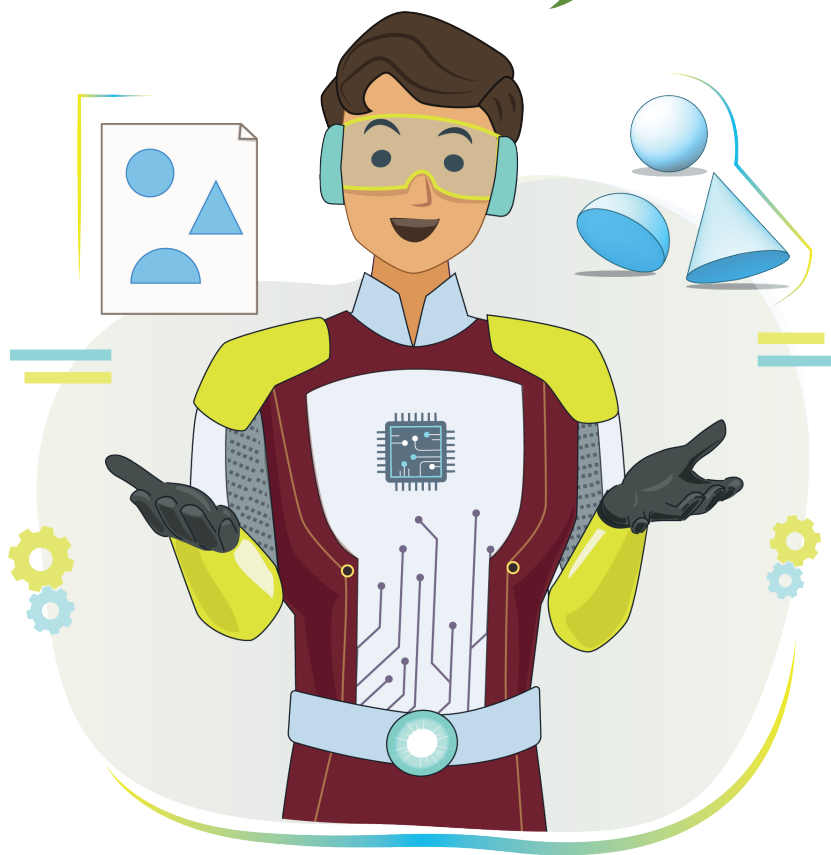


3D Environment

01

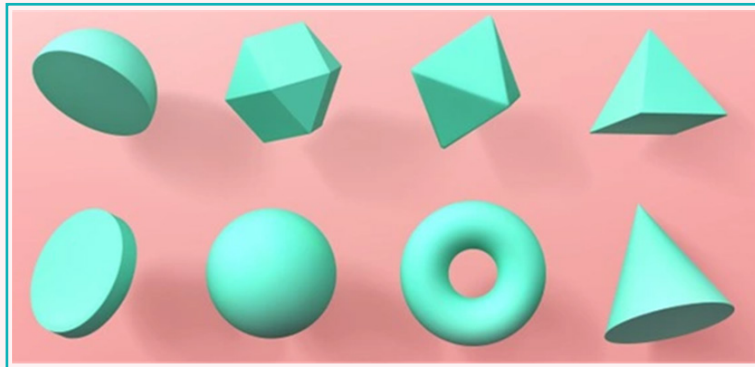
Have you ever wondered why shapes look different on a piece of paper and in real life?

Let's see why that happens!



All around us, there are many different objects. We are surrounded by pens, notebooks, bottles, balls, televisions, computers, etc. All these objects are of various shapes and sizes.

All the shapes that we see around us are 3-Dimensional (3D). 3D shapes are also called solid shapes. Length, width, and height (or depth or thickness) are the three dimensions of a 3D shape. Shapes whose dimensions are only length and height are 2D shapes.



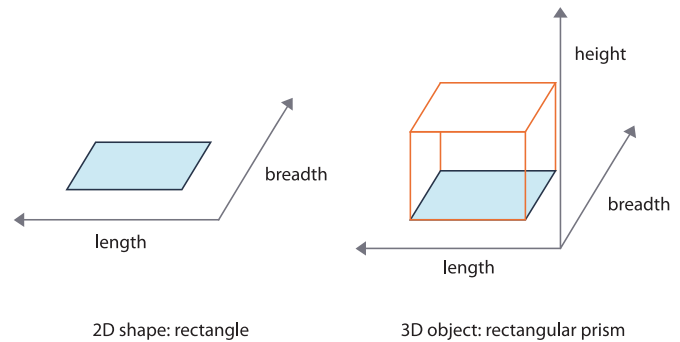
We mostly use such shapes in Geometry. Geometry is a practical section of mathematics which works with the shapes and sizes of different figures and their properties. Geometry can be divided into two types: plane and solid geometry. Plane geometry deals with flat shapes like lines, curves, polygons, etc., which can be drawn on a piece of paper. Solid geometry involves three-dimensional shapes like cylinders, cubes, and spheres which can be created in specialised applications like Autodesk Maya, Blender, Fusion 360, Tinkercad etc.

Activity

Here is a table that mentions names of 3D shapes and their 2D counterparts. Complete it with appropriate examples.

2D	3D	Example
Circle	Sphere	
Triangle	Prism	
Square	Cube	
Rectangle	Rectangular Cuboid	

Now, let's look at how to convert a 2D shape into a 3D shape and which are the tools we need to use.



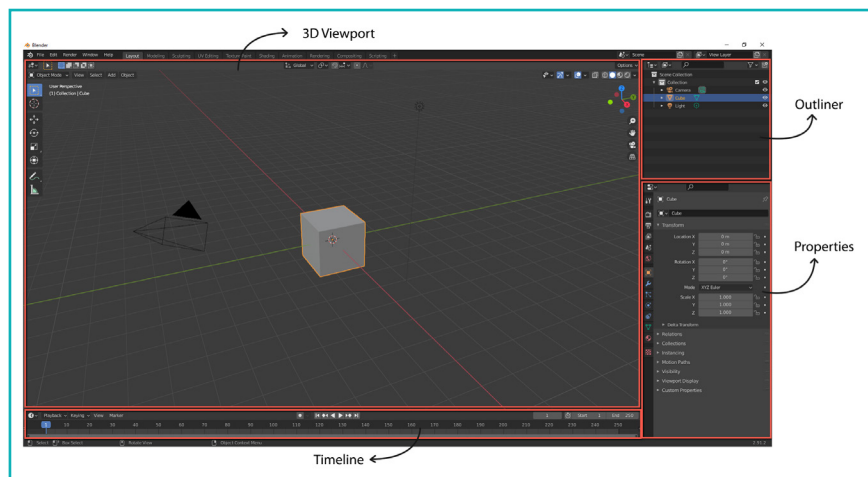
We have several options to choose from when working with 3D shapes. Blender is one such application which is open source and packed with features.

Note:

Blender is a 3D computer graphics software used to create animated films, 3D models and computer games.

The **Blender** interface consists of several windows. Its default starts up shows the “Layout” workspace in the main area. This is a general workspace to preview scenes and objects. It includes the following Editors:

- 3D Viewport on the top left
- Outliner on the top right
- Properties on the bottom right
- Timeline on the bottom left



Workspaces are essentially predefined window layouts. Blender allows us to create customised workspaces for different tasks like modeling, animating, and scripting.

Exercise

State whether True or False

1 A Ball is an example of a sphere.

 _____

2 Plane geometry deals with flat shapes like lines, curves, polygons, etc., that can be drawn on a piece of paper.

 _____

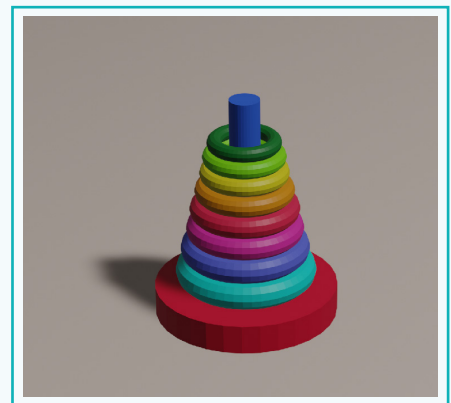
3 Blender is used to create animated movies.

 _____

We will explore each function of Blender as we go ahead.

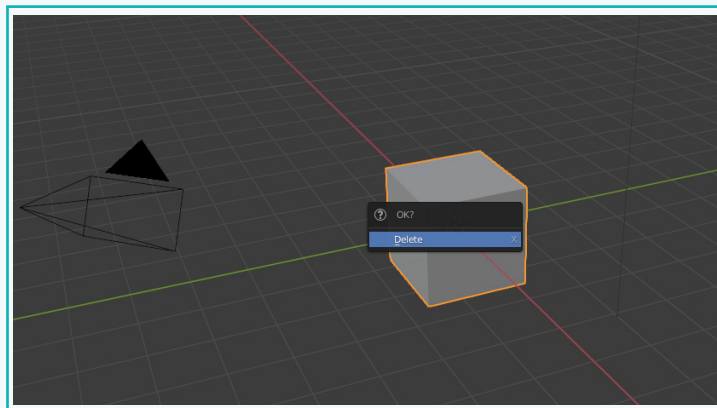
Activity

Let's work on a small activity and create a toddler's toy. Observe it carefully. The shapes it has include a cylindrical rod and spherical rings.



Activity

- 1 By default, when we start the application, we have a cube, a camera and light in the viewport. We will delete it and create a new object. To delete an object, select it and press Delete on the keyboard.



3D viewport: The main window in which 3D objects are created, edited, and animated for a scene is 3D viewport. It is used to interact with the 3D scene for a variety of purposes, like modeling, animating, texture painting, etc.

The viewport displays a grid floor. The lines that form the grid also consist of red, green, and blue lines which represent the world coordinate axes X, Y and Z respectively.

Being a 3D application, Blender lets us view an object from all angles. There are multiple ways for navigation within the viewport. We will learn a method that uses a mouse.

Orbit: To rotate the view, click and drag the Middle Mouse Button (MMB) to the viewport's area.

Pan: Moving up, down, left and right is termed panning movement. To pan the view, hold down the shift key and drag the (MMB) to the 3D Viewport.

Zoom: Moving the camera forward and backward is a zoom movement. Hold down Ctrl and drag the MMB to zoom in and out.

Note:

MMB stands for Middle Mouse Button

Hotkeys (shortcut keys) for navigation:

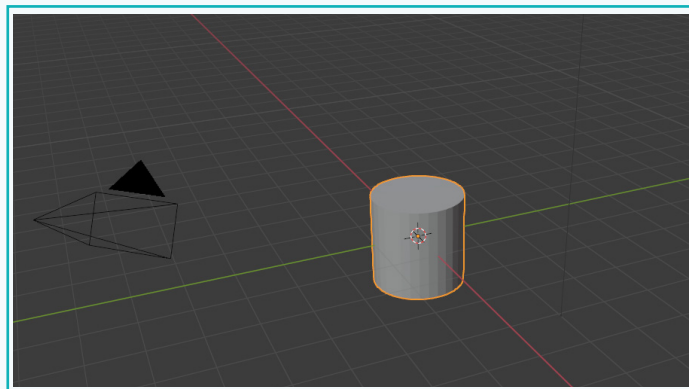
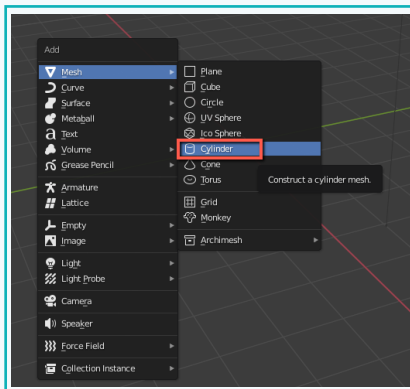
Navigation	Hotkey
Orbit	Drag MMB
Pan	Shift + MMB
Zoom	Ctrl + MMB

Activity

Most hotkeys affect the active area (the one that is in focus), so confirm that the cursor is in the correct area before using hotkeys.

Now that we have an empty scene, let us try to add some elements in it. We will create the rod first.

- 2 To add an object, press Shift + A on the keyboard. Select a type of Mesh to create. Here we will use a cylinder.

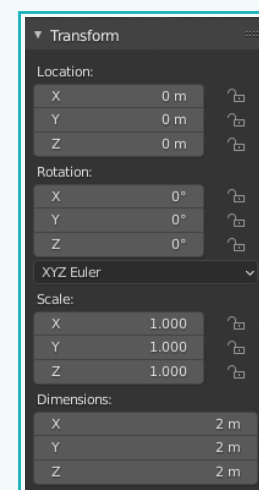


- 3 To ensure that it looks like the object, we need to size it appropriately. There are multiple ways to size an object. In this case, we will use the sidebar.

Note:

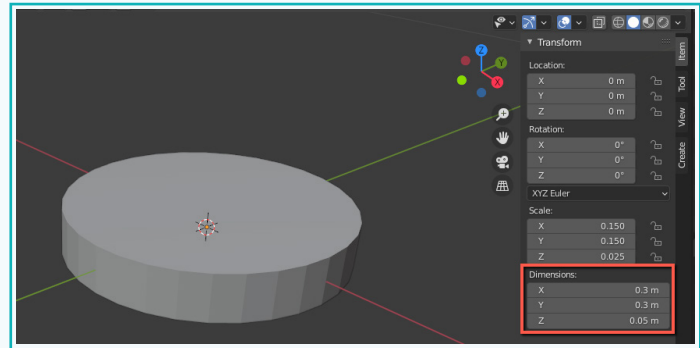
The Sidebar (on the right side of the editor area) contains panels with object settings within the editor. It also contains the editor. The N key toggles the sidebar's visibility.

The sidebar displays the transform parameters of the selected object. These include location, rotation (orientation), scale, and dimensions.

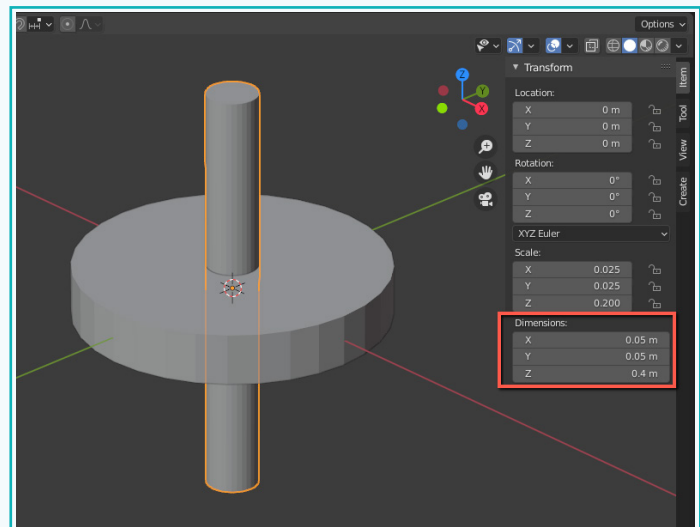


Activity

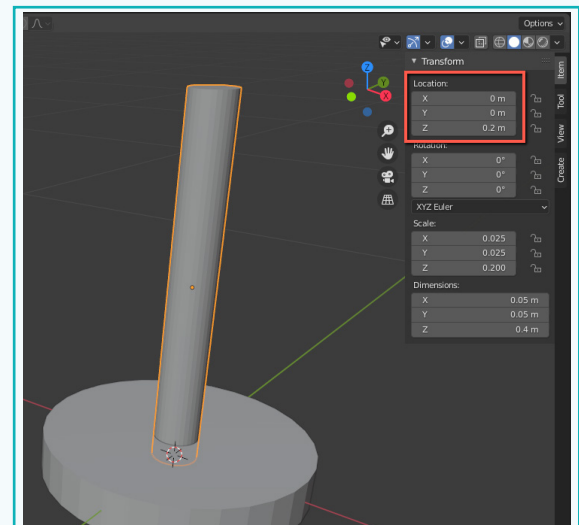
- 4 To size it correctly, we need to alter its dimensions.



- 5 For the stem of the toy, we will add another cylinder and set the dimensions as X: 0.05m, Y:0.05m, Z: 0.4m.

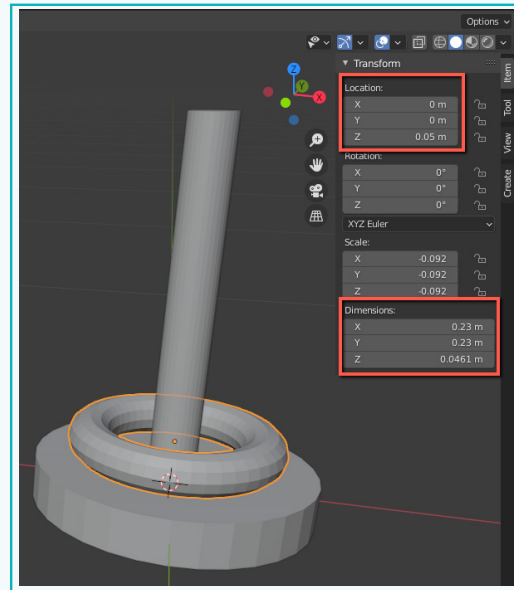
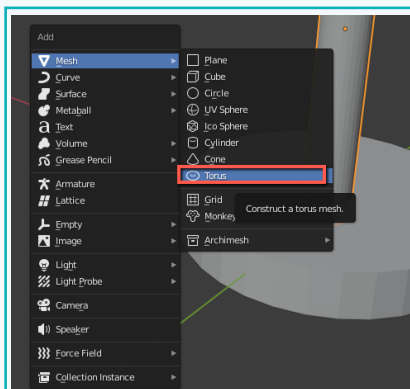


Our aim is to create the cylindrical base of the toy, which is at the bottom. Right now, it is in the middle. To place the object correctly, we need to change its location in the 3D space. In a 3D environment, we can identify an object's location using the Cartesian Coordinate System. Each point in 3D space is defined with respect to 3 axes.

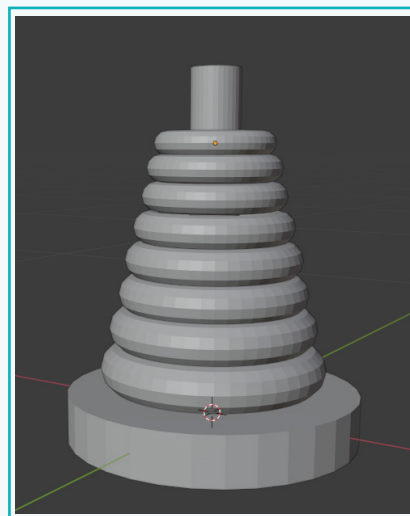


Activity

Using the same technique, we will add the stackable rings using a torus mesh.



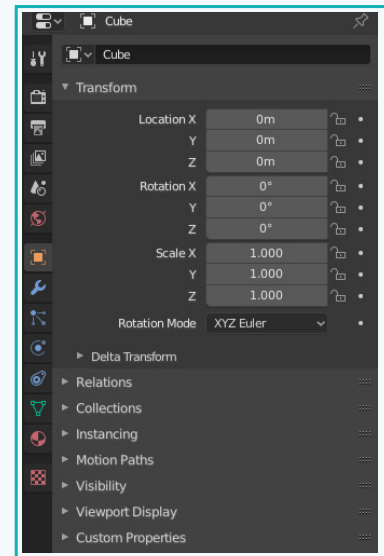
- 6 Use the torus mesh to stack the rings and add a mesh plane to represent the floor.



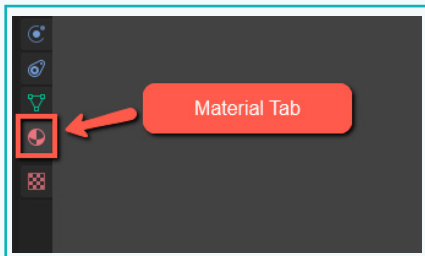
Children's toys are made of plastic. To give our object a plastic look, we will apply a suitable material to it. The appearance of an object is controlled by the material on it. Materials define the substance the object is made of, its colour and texture, and how light interacts with it. They can be created in the material properties.

Activity

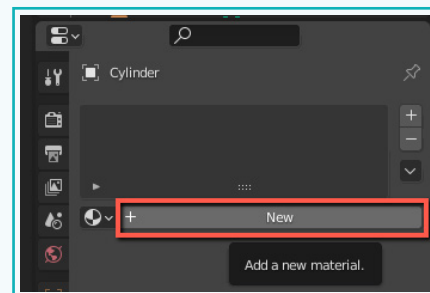
The properties panel displays and allows us to edit active data, including the active scene and object. The properties panel has several categories which can be chosen via tabs (the icons in the column to its left). Each tab regroups properties and settings of a data type.



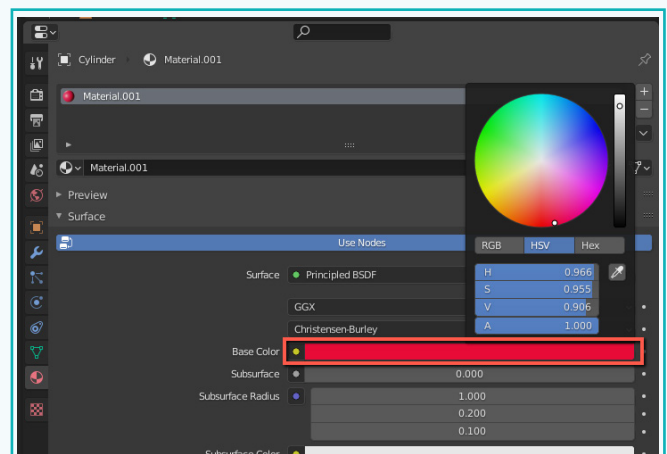
- 7 To change the material, switch to the materials tab in the properties panel.



- 8 Select the object in the viewport and click on New to create a new material.

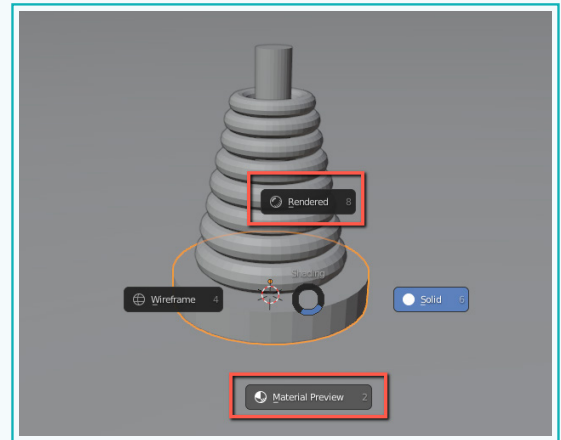


Select a colour in the base colour tab.

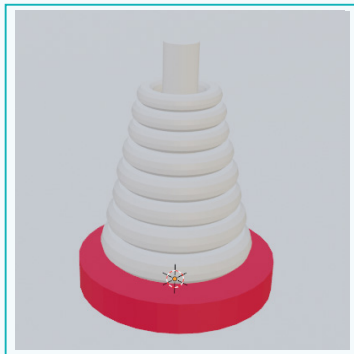


Activity

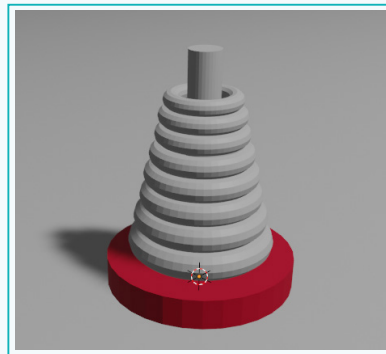
This applies the material to the selected object. If the material is not visible in the viewport, press “Z” and set the viewing mode to material preview or rendered view.



Material view



Rendered view



We can create materials to suit any object.



In this lesson, we learnt the difference between 2D and 3D objects. We also understood how to convert 2D objects into 3D. Many software exist for this; we explored Blender. We used Blender to create a 3D version of a children's toy.

Exercise

Answer in one line

4 What is 3D viewport?



Tick mark ☒ the correct answer

5 Which of the following is the correct pair?

a. ☐

Navigation

Orbit

Hotkey

Drag MMB

b. ☐

Pan

Ctrl + MMB

c. ☐

Zoom

Shift + MMB

6 Material does not define the _____ of an object.

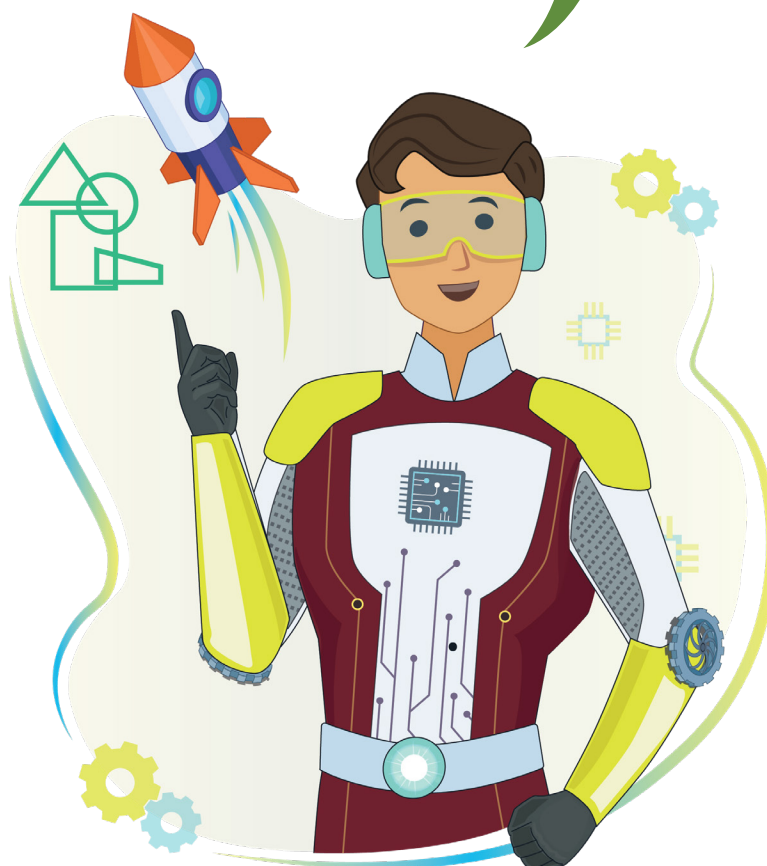
a. Substance ☐

b. Colour ☐

c. Size ☐

d. Texture ☐

We can combine different shapes to create a single object.



Tech Flash

- **Workspaces** : Workspaces are predefined window layouts.
- **Sidebar** : The sidebar contains panels with object settings within the editor, and the editor itself.
- **Important hotkeys (shortcut keys)** :

Shift + A	Add object
N	Toggle sidebar
Z	Change view mode